

Problem 65 (Italy MO 2019 Problem 2). Let p and q be prime numbers such that $p + q^2$ is a perfect square. Prove that $p^2 + q^n$ is not a perfect square for any positive integer n .

Solution. Let $p + q^2 = a^2$. This implies

$$p = a^2 - q^2 = (a - q)(a + q).$$

As p is a prime, we must have $a - q = 1$ and $a + q = p$. Thus, $2q = p - 1$.

Suppose on the contrary that $p^2 + q^n = b^2$ for some positive integer b . Then we have

$$q^n = b^2 - p^2 = (b - p)(b + p).$$

Let $d = (b - p, b + p) = (b - p, 2p)$. Note that d also divides q^n . As $p \neq q$ (from $2q = p - 1$), this yields $d = 1, 2$.

If $d = 1$, we must have $b - p = 1$ and $b + p = q^n$. Thus, $2p = q^n - 1$, and hence $2(2q + 1) = q^n - 1$. This implies $2 \equiv -1 \pmod{q}$, and hence $q = 3$. Then $p = 2q + 1 = 7$. But then $3^n = q^n = 2p + 1 = 15$ has no solution.

If $d = 2$, q must be 2, and so $p = 2q + 1 = 5$. The equation becomes $2^n = (b - 5)(b + 5)$. So $b - 5$ and $b + 5$ are powers of 2 with difference 10. Since $v_2(10) = 1$, we must have $b - 5 = 2$. But then $b + 5 = 12$ is not a power of 2, contradiction.

So $p^2 + q^n$ is not a perfect square for any positive integer n . □

Problem 66. Let n be a positive integer. Let A be the number of tuples $(a_1, a_2, \dots, a_r)$ of positive integers such that $a_1 + a_2 + \dots + a_r = n$, $a_{r-1} > a_r$ and $a_i > a_{i+1} + a_{i+2}$ for $i = 1, 2, \dots, r - 2$ (where r can be any positive integer). Define the sequence $\{x_n\}$ by $x_1 = 1$, $x_2 = 2$ and $x_n = x_{n-1} + x_{n-2} + 1$ for $n \geq 3$. Let B be the number of tuples $(b_1, b_2, \dots, b_s)$ of positive integers such that $b_1 + b_2 + \dots + b_s = n$, $b_1 \geq b_2 \geq \dots \geq b_s$, and there exists k for which the set $\{b_1, b_2, \dots, b_s\}$ is the same as the set $\{x_1, x_2, \dots, x_k\}$ (where s can be any positive integer). Prove that $A = B$.

Solution. We show that there is a bijection between tuples $(a_1, a_2, \dots, a_r)$ counted in A and tuples $(b_1, b_2, \dots, b_s)$ counted in B .

For each tuple $(a_1, a_2, \dots, a_r)$, define $(b_1, b_2, \dots, b_s)$ to be the tuple of decreasing positive integers such that there are a_r copies of x_r , $a_{r-1} - a_r$ copies of x_{r-1} , and $a_j - a_{j+1} - a_{j+2}$ copies of x_j for $j = r - 2, r - 3, \dots, 1$. In view of the given inequalities on $\{a_i\}$, we must have $\{b_1, b_2, \dots, b_s\} = \{x_1, x_2, \dots, x_r\}$. Also, we have

$$\sum_{i=1}^s b_i = \sum_{j=1}^{r-2} (a_j - a_{j+1} - a_{j+2})x_j + (a_{r-1} - a_r)x_{r-1} + a_r x_r = x_1 a_1 + (x_2 - x_1)a_2 + \sum_{j=3}^r (x_j - x_{j-1} - x_{j-2})a_j = \sum_{j=1}^r a_j = n.$$

So this gives a tuple counted in B .

This mapping is invertible. For each tuple $(b_1, b_2, \dots, b_s)$ where $\{b_1, b_2, \dots, b_s\} = \{x_1, x_2, \dots, x_r\}$, if there are c_j copies of x_j for $j = 1, 2, \dots, r$, we define $a_r = c_r$, $a_{r-1} = c_{r-1} + a_r$ and $a_j = c_j + a_{j+1} + a_{j+2}$ for $j = r - 2, r - 3, \dots, 1$. Then $a_r > 0$, $a_{r-1} - a_r = c_{r-1} > 0$ and $a_j - a_{j+1} - a_{j+2} = c_j > 0$ for $j = 1, 2, \dots, r - 2$. Also, in view of the definition, it is clear that this mapping is the inverse of the above mapping, and so $\sum_{j=1}^r a_j = \sum_{i=1}^s b_i = n$. This establishes the bijection, and so $A = B$. □

Problem 67 (ARMO 2017 Grade 11 Problem 3). Let $a_1, a_2, \dots, a_n$ be given positive real numbers. Prove that there exist positive real numbers $b_1, b_2, \dots, b_n$ such that $b_i \geq a_i$ for each i , $\frac{b_i}{b_j}$ is an integer whenever $b_i \geq b_j$, and $b_1 b_2 \dots b_n \leq 2^{\frac{n-1}{2}} a_1 a_2 \dots a_n$.

Solution. For each $m = 1, 2, \dots, n$, we consider the sequence $\{b_1^{(m)}, b_2^{(m)}, \dots, b_n^{(m)}\}$ defined by $b_i^{(m)} = 2^{k_{im}} a_m$, where k_{im} is the smallest integer such that $2^{k_{im}} a_m \geq a_i$ (equivalently, $k_{im} = \left\lceil \log_2 \frac{a_i}{a_m} \right\rceil$). By definition, we have $b_i^{(m)} \geq a_i$ for each i , and

$$\frac{b_i^{(m)}}{b_j^{(m)}} = \frac{2^{k_{im}} a_m}{2^{k_{jm}} a_m} = 2^{k_{im} - k_{jm}} \text{ is an integer whenever } b_i^{(m)} \geq b_j^{(m)}.$$

$$b_1^{(m)} b_2^{(m)} \dots b_n^{(m)} \leq 2^{\frac{n-1}{2}} a_1 a_2 \dots a_n.$$

Now,

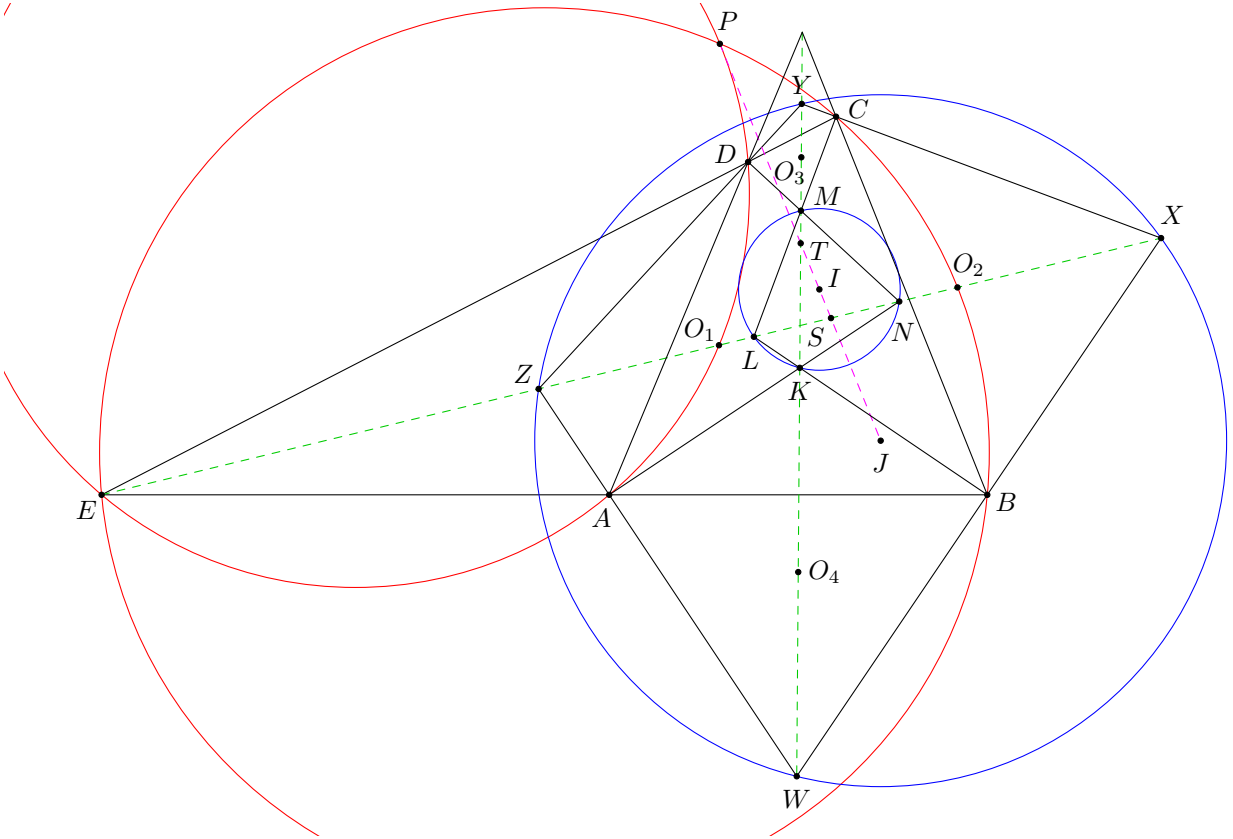
$$\prod_{m=1}^n b_1^{(m)} b_2^{(m)} \cdots b_n^{(m)} = \prod_{i=1}^n \prod_{m=1}^n b_i^{(m)} = \prod_{i=1}^n \prod_{m=1}^n 2^{k_{im}} a_m = 2^{\sum_{i,m} k_{im}} (a_1 a_2 \cdots a_n)^n.$$

Note that $k_{ii} = 0$ and $k_{im} + k_{mi} = \left\lceil \log_2 \frac{a_i}{a_m} \right\rceil + \left\lceil \log_2 \frac{a_m}{a_i} \right\rceil = 0, 1$ if $i \neq m$ since $\log_2 x + \log_2 \frac{1}{x} = 0$. Therefore, we have

$$\prod_{m=1}^n b_1^{(m)} b_2^{(m)} \cdots b_n^{(m)} \leq 2^{\frac{n(n-1)}{2}} (a_1 a_2 \cdots a_n)^n.$$

By the pigeonhole principle, there exists m such that $b_1^{(m)} b_2^{(m)} \cdots b_n^{(m)} \leq 2^{\frac{n-1}{2}} a_1 a_2 \cdots a_n$ as desired. $\square$

Problem 68 (Brazil MO 2016 Problem 6). Let $ABCD$ be a convex quadrilateral whose sides are not parallel and whose vertices are not concyclic. The lines AB and CD meet at E . The circumcircles of $\triangle ADE$ and $\triangle BCE$ meet again at P . The four internal angle bisectors of $ABCD$ enclose a cyclic quadrilateral with circumcentre I . The four external angle bisectors of $ABCD$ enclose a cyclic quadrilateral with circumcentre J . Prove that P, I, J are collinear.



Solution. We only work on the configuration as shown. Let the internal angle bisectors meet at K, L, M, N as shown, and let the external angle bisectors meet at W, X, Y, Z as shown. Since Z and N are the incentres and excentres of $\triangle ADE$, and L and X are the incentres and excentres of $\triangle BCE$, the points E, Z, L, N, X are collinear. Similarly, Y, M, K, W are collinear. Let S and T be the intersection points of IJ with ZX and YW . Also, let O_1 and O_2 be the midpoints of ZN and XL respectively. Note that O_1 is the centre of $(NDZA)$ since $\angle ZDN = \angle ZAN = 90^\circ$. Similarly, O_2 is the centre of $(LCXB)$.

We first show that S is a centre of similitude between $(KLMN)$ and $(WXYZ)$. As I and J are the centres of the circles and S lies on IJ , it suffices to show $IN \parallel JZ$, for example. We check that

$$\begin{aligned} \angle INZ &= \angle INL = 90^\circ - \angle LMN = 90^\circ - \angle CMD = \frac{C}{2} + \frac{D}{2} - 90^\circ, \\ \angle JZN &= \angle JZX = 90^\circ - \angle XWZ = 90^\circ - \frac{A}{2} - \frac{B}{2} = \angle INZ. \end{aligned}$$

This proves our claim.

It is well known that O_1 lies on $(PEAD)$ as it is the midpoint of the incentre and excentre of $\triangle EAD$. Similarly, O_2 lies on $(PEBC)$. Note that P is the centre of spiral similarity sending AD to BC . Thus, it also maps the midpoint O_1 of arc AD to the midpoint O_2 of arc BC . It follows that

$$\frac{PO_1}{PO_2} = \frac{O_1D}{O_2C} = \frac{O_1N}{O_2L}.$$

Next, the homothety with centre S maps N to Z and maps L to X . So it also maps the midpoint O_1 of NZ to the midpoint O_2 of LX . This implies

$$\frac{SO_1}{SO_2} = \frac{O_1N}{O_2L} = \frac{PO_1}{PO_2}.$$

Therefore, PS is the internal angle bisector of $\angle O_1PO_2$. Similarly, PT is the internal angle bisector of $\angle O_3PO_4$, where O_3 and O_4 are the midpoints of MY and KW respectively. If we can show that the two internal angle bisectors are the same, then PST is this angle bisector, and hence P, I, J are collinear.

This can be proved by some angle chasing. Indeed, we have

$$\angle O_1PO_3 = \angle O_1PD + \angle O_3PD = \frac{1}{2}\angle APD + \frac{1}{2}\angle CPD = \frac{1}{2}\angle APC = \frac{1}{2}\angle APB + \frac{1}{2}\angle CPB = \angle O_4PB + \angle O_2PB = \angle O_4PO_2.$$

This easily implies $\angle O_1PO_2$ and $\angle O_3PO_4$ have the same internal angle bisectors as desired. □